



Responses of soil organic carbon, microbial biomass carbon, and microbial quotient to ground cover management practices in Chinese orchards: a data synthesis

Weiting Ding · Jinmei Chen · Yeru Wu · Jianuo Mu · Zhidong Qi · Francis Zvomuya · Hailong He

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Abstract

Background and aims Soil microbial quotient (SMQ), the ratio of soil microbial biomass carbon (SMBC) to organic carbon (SOC), is considered a key predictor of soil quality. However, the variations in SMBC and SMQ under ground cover management (GCM) in orchard systems remain poorly understood. **Methods** This study established a national database of SOC, SMBC, and SMQ, as well as associated climatic and edaphic factors, and agronomic practices from 288 field experiments across China. We leveraged the data to assess the responses and controlling factors of SOC, SMBC, and SMQ to GCM practices using a meta-analysis.

Results Our findings demonstrate that GCM significantly increases SOC (20%), SMBC (35%), and SMQ (19%) compared to conventional clean (bare ground) tillage. The categorical meta-analyses reveal that the

response varies with management and environmental factors. The effect of GCM on SOC, SMBC, and SMQ is prominent in tropical rainy areas with low SOC content ($< 10 \text{ g kg}^{-1}$), neutral soils ($\text{pH}=6\text{--}8$), full cover with live grass, and shallow soils. Furthermore, the random forest modeling further demonstrates that soil depth is the dominant driver for responses of SOC, SMBC, and SMQ to GCM. The GCM-induced changes in SOC, SMBC and SMQ decline significantly as soil depth increased. The significant increase in SMQ was mainly observed in the upper 40 cm soil layer, which indicates that the ability of microorganisms to assimilate carbon decreases with soil depth.

Conclusion Overall, we conclude that GCM practices could improve carbon sink and boost biological soil health in Chinese orchards. The response, however, varied depending on local environmental factors. We also highlight that a careful assessment of the site-specific conditions is needed to determine the appropriate GCM practices for potential benefits.

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W. Ding · J. Chen · Y. Wu · J. Mu · Z. Qi · H. He (✉)
College of Natural Resources and Environment, Northwest A&F University, Yangling, Shaanxi 712100, China
e-mail: hailong.he@hotmail.com

F. Zvomuya · H. He
Department of Soil Science, University of Manitoba,
Winnipeg, MB R3T 2N2, Canada

Highlights

- Ground cover management (GCM) overall increases soil organic carbon (SOC), microbial biomass carbon (SMBC), and microbial quotient

(SMQ) by 19–35% compared to clean tillage in orchards.

- GCM effect on microbial properties varies with management practices and environmental factors.
- Soil depth is the dominant driver of SOC, SMBC, and SMQ responses to GCM
- GCM is a promising strategy for increasing SOC content in orchards.

Keywords Orchard · Cover crops · Microbial biomass pools · Microbial quotient · Meta-analysis

Introduction

Soil represents the largest carbon pool in terrestrial ecosystems, with carbon storage surpassing that of both vegetation and the atmosphere combined (Jobbágy and Jackson 2000). It actively sequesters atmospheric CO₂ while simultaneously functioning as a dynamic carbon source (Lal 2004). Enhancing soil organic carbon (SOC) sequestration is regarded as a proven and economically viable natural strategy for reducing atmospheric CO₂ concentration and mitigating global warming (Rumpel et al. 2018; Xu et al. 2022). Agricultural practices such as conservation tillage, cover crop (CC), and biochar application have recently been designated as climate-smart agricultural practices (Lipper et al. 2014). These practices aim to sequester more carbon in soils by increasing carbon inputs and reducing carbon outputs (Bai et al. 2019). Among these practices, cover cropping, which involves cultivating herbaceous plants between woody crop rows (de Torres et al. 2021), is recognized as a cost-effective and highly efficacious agricultural practice for enhancing SOC (Guenet et al. 2021; Ryan 2021; Zhang et al. 2022b).

Orchard cultivation has emerged as a crucial component of agriculture (Ren et al. 2023). Optimizing orchard management offers significant potential for promoting SOC sequestration (Wei et al. 2017; Zhao et al. 2021). Clean tillage (CT) and ground cover management (GCM) are two contrasting and representative practices in orchard management (Xiang et al. 2022). The traditional practice of CT in orchards involves maintaining the soil bare between the rows and/or alleyways of tree crops through regular plowing and/or spraying herbicides (Xiang et al.

2022; Xie et al. 2022), which increases the risk of soil erosion and depletes SOC (Zhao et al. 2021). Ground cover management (GCM) is one of the most important agricultural practices, referring to the occupation of field soil between the rows and/or alleyways and under the crop canopy (Shaxson 1999). GCM can be achieved through several agro-ecological practices, utilizing either living cover (e.g., cultivated or natural cover crops) or dead cover (e.g., straw, crop residue, twigs, stones, and plastic mulch) (Mottes et al. 2014) to reduce weed growth, soil and nutrient loss, and improve land productivity (Gentil-Sergent et al. 2022).

Over the last few decades, GCM has been widely adopted in developing countries (Li et al. 2021b; Ma et al. 2021). China has the world's largest orchard area and is one of the leading fruit producers globally (Xiang et al. 2022), which presents significant opportunities for SOC storage (Zhang et al. 2022b). However, there is large variation in SOC sequestration in orchard GCM practices (Hu et al. 2022). For instance, studies have shown that GCM can promote SOC sequestration (Liu et al. 2013; Qian et al. 2015; Wei et al. 2018; Hu et al. 2023), while a study conducted in citrus orchards found that SOC loss was significantly reduced by 98% under GCM (Zheng et al. 2021b). However, Wang et al. (2020) observed higher SOC concentration under CT than under grass planting in a jujube (*Ziziphus jujuba* Mill) orchard. Cui et al. (2015) observed that CC did not significantly impact soil organic matter (SOM) in a guava orchard. Such variability in SOC accumulation or loss could be related to variations in regional climate, soil characteristics, and agronomic practices at experimental sites, and it may change significantly over time (Banger et al. 2010; Schmidt et al. 2011; Poelplau and Don 2015; Payen et al. 2021). This is possibly attributed to the complexity of agricultural systems, the interactions among multiple environmental factors and site-specific GCM practices, which play an important role in controlling the input of below-ground biomass and above-ground plant residues, as well as microbial and physico-chemical processes (Barman et al. 2014; Ren et al. 2019). Recently, a meta-analysis was conducted to quantitatively assess the effect of orchard grass coverage on SOC stocks across China (Xiang et al. 2022), which enhances our understanding of the CC-driven magnitude and variability in the response of SOC stocks. However,

the limited number of studies and the exclusion of soil depth below 30 cm have resulted in uncertainty regarding the extent and drivers of SOC sequestration. In addition, SOC accumulation is closely linked to changes in microbial characteristics, such as soil microbial biomass carbon (SMBC) and soil microbial quotient (SMQ, the SMBC-to-SOC ratio) (Cao et al. 2021). However, few studies have attempted to clarify the response of these specific soil microbial properties to GCM practices on a large scale. Therefore, this issue warrants further investigation to advance our understanding of SOC accumulation within GCM systems.

Soil microorganisms play a pivotal role in regulating SOC dynamics (Liang et al. 2011; Zhang et al. 2017; Sun et al. 2020), and organic C components from microorganisms are necessary for enhancing SOC (Liang et al. 2017). Soil microbial biomass is a responsive index that reflects the key biochemical dynamics and processes of soil and has an important influence on soil turnover (Ren et al. 2019; Li et al. 2021c). As a generally accepted indicator of soil health, SMQ represents a biologically active fraction of SOM (Bachar et al. 2010; Zhou et al. 2017). Therefore, SMQ could be further linked to SOC turnover, which is mediated either by carbon inputs, the quality of carbon substrates, or both (Coban et al. 2022). However, a quantitative comparison of SMBC and SMQ and their controls in orchard systems on regional scales (e.g., China) has not been reported, and therefore requires further elucidation at regional and national scales. A greater understanding of SMBC and SMQ and their controlling factors is essential for comprehending the carbon cycle and microbial biogeography (Martiny et al. 2006).

Here, we conducted a meta-analysis to evaluate the impact of orchard GCM on SOC, SMBC, and SMQ. This study aimed to (1) summarize and characterize the effects of GCM on SOC, SMBC, and SMQ in orchards across China, within the 0–100 cm soil layer; and (2) examine the extent to which GCM contributes to SOC accumulation and how the benefits of GCM on SOC, SMBC, and SMQ vary in relation to different control factors. The hypothesis to be tested was that GCM would overall increase SOC, SMBC, and SMQ relative to CT, and we further hypothesize that the effects of GCM on SOC, SMBC, and SMQ will be affected by the climatic factors, edaphic

variables, and agronomic practices at the experimental sites.

Materials and methods

Study selection

Research papers published during 1990–2021 in the Web of Science and China National Knowledge Infrastructure databases were selected. The search keywords included China, orchard, fruit, cover crop, ground cover, ground management, catch crop, mulch, clean tillage, bare land, traditional tillage, soil organic carbon, soil organic matter, microbial biomass carbon, and microbial quotient.

Relevant publications were screened according to the following criteria: (1) only field experiments conducted in China were included; (2) treatments had three or more replications; (3) SOC or SMBC or SMQ should be clearly presented; and (4) only data from the mulch (cover crop or ground cover) and control (clean tillage and maintain the soil bare) treatments were extracted from the multifactor experiments. A total of 288 publications that met these criteria were included (Fig. 1 and Table S1).

For each study, the longitude, latitude, elevation, mean annual temperature (MAT), mean annual precipitation (MAP), edaphic factors (SOC, TN content, pH, and soil sampling depth), experimental period, tree age, the type and pattern of cover, the source and species of grass, mowing, and placement of mowed residues were recorded. Data for SOC, SMBC, and SMQ were extracted from relevant publications. If the data were presented graphically, WebPlotDigitizer was used to extract the data (Burda et al. 2017).

A total of 15 explanatory variables were identified and categorized into three groups as agronomic practices, climatic, and edaphic variables, and (Table S2). We followed thresholds derived from previous meta-analyses (Hu et al. 2022; Liu et al. 2022; Xiang et al. 2022), as well as the general level breakpoints of the dataset. Basic details of the 288 publications are summarized in Table S1.

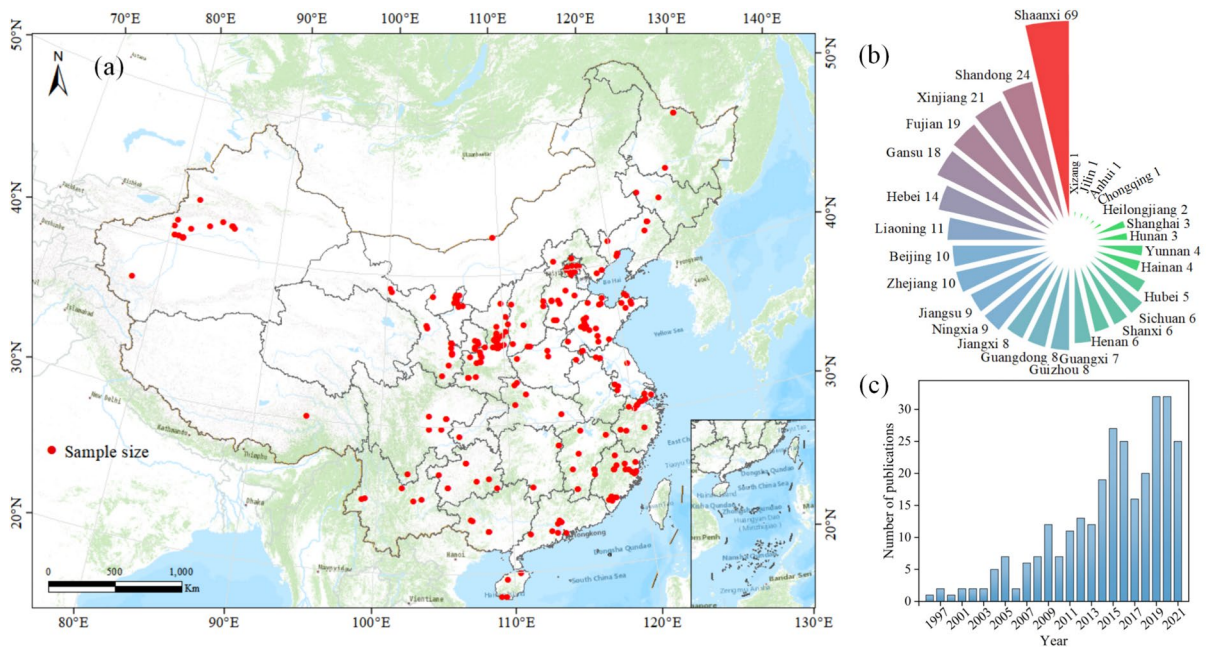


Fig. 1 Distribution of the 288 experimental sites across China compiled in our study (a), site distribution by province (b), and annual publication volume (c)

Meta-analysis

The natural logarithm response ratio ($\ln RR$) of GCM and CT was calculated using Eq. (1) (Hedges et al. 1999):

$$\ln RR = \ln \frac{\bar{x}_{GCM}}{\bar{x}_{CT}} \quad (1)$$

where $\bar{x}_{GCM}$ and $\bar{x}_{CT}$ are the means of GCM and CT, respectively.

Percentage change in SOC, SMBC, and SMQ as follows Eq. (2):

$$(e^{\ln RR_{++}} - 1) \times 100\% \quad (2)$$

where the weighted response ratio ($\ln RR_{++}$) was calculated as Eq. (3):

$$\ln RR_{++} = \frac{\sum_{i=1}^n \sum_{j=1}^m w_{ij} \ln RR_{ij}}{\sum_{i=1}^n \sum_{j=1}^m w_{ij}} \quad (3)$$

where n and m represent the number of groups compared and comparisons in the corresponding group. The weight (w_{ij}) of each study was estimated by Eq. (4):

$$w_{ij} = 1/v \quad (4)$$

where v is the variance and calculated by Eq. (5):

$$v = SD_{GCM}^2 / (n_{GCM} \times x_{GCM}^2) + SD_{CT}^2 / (n_{CT} \times x_{CT}^2) \quad (5)$$

where SD represent the standard deviation, n is the independent sample size. When a publication reported only the standard error (SE), SD was estimated by Eq. (6):

$$SD = SE \sqrt{n} \quad (6)$$

where n is replicate numbers. In situations where only the mean value was reported without the accompanying SE or SD values, SD was calculated as 0.1 times of the means (Luo et al. 2006).

95% confidence interval (CI) was estimated using Eq. (7)

$$95\% \text{ CI} = (\ln RR_{++}) \pm 1.96 \text{ SE} (\ln RR_{++}) \quad (7)$$

The 95% CI was calculated using bootstrapping tests (4999 iterations) (Zhao et al. 2020b). The significance of the orchard cover effect was determined by examining whether the 95% CI overlap with 0.

MetaWin 2.1.3 was used to perform a meta-analysis (Rosenberg et al. 1997). The significance of the total heterogeneity among studies (Q_T) and between studies (Q_b) was assessed using the chi-square test. An I^2 index above 75% indicates a significant amount of heterogeneity (Zhao et al. 2020a). Test results for data heterogeneity were summarized in Table S3. All $\ln RR$ values were tested for their fit to the Gaussian (normal) distribution function (Figure S1). Publication bias was tested using Rosenthal's fail-safe number (R), a number greater than $5n + 10$ (where n is the sample size) indicates the absence of publication bias (Rosenberg 2005).

Statistical analysis

A random forest (RF) model was used to assess the importance of each explanatory variable within a given dataset. The relative contribution of factors (i.e., climate and edaphic variables, and management practices) in explaining the variability in SOC, SMBC, and SMQ was quantitatively determined using the “randomForest” package in R (version 3.4.2). Finally, a sample linear regression analysis was used to examine relationships between the $\ln RR$ of SOC, SMBC, and SMQ and the most critical drivers (MAT, MAP, duration, and initial SOC and TN).

Results

Data overview and overall responses of SOC, SMBC, and SMQ to GCM practices

Our dataset comprised 288 independent research sites in China, where orchard GCM experiments were conducted (Fig. 1a). These sites spanned 28 provinces, with Shaanxi ($n=69$), Shandong ($n=24$), and Xinjiang ($n=21$) being the top three provinces in terms of the number of sites (Fig. 1b). The annual publication volume exhibited a fluctuating upward trend from 1993 to 2021, indicating the growing acceptance and implementation of cover crop practices in China (Fig. 1c). The $\ln RR$ of SOC, SMBC, and SMQ followed a normal distribution (Figure S1). The I^2 index (98%–99.9%) and Q_T values (263–3589, $p < 0.001$) indicated a high overall heterogeneity of the data (Table S3). These indicated that our data met the criteria for conducted a meta-analysis.

GCM significantly increased SOC, SMBC, and SMQ by 20%, 35%, and 19%, respectively ($p < 0.05$, Fig. 2a). The changes induced by GCM in each variable consistently showed a positive response compared to continuous CT (Fig. 2a). Most of the published studies focused on regions located between 30° and 40° of north latitudes, revealing an average SOC, SMBC, and SMQ of approximately 7 g kg⁻¹, 231.1 mg kg⁻¹, and 2.5%, respectively. Notably, the highest (9.5 g kg⁻¹) and lowest (6 g kg⁻¹) levels of SOC contents, were observed between latitudes of 20–30° and 30–40°, respectively. The highest levels of SMBC (482.7 mg kg⁻¹) and SMQ (4.5%) were observed in high-latitude regions (40–50°), while the lowest levels of SMBC (87.5 mg kg⁻¹) and SMQ (1%) were found in low-latitude regions (< 20°) (Fig. 2b–d).

Response of SOC, SMBC, and SMQ to GCM under various agronomic practices

SOC content was significantly greater under GCM relative to the control regardless of agronomic practice ($p < 0.05$) (Fig. 3 and Table S4). GCM significantly increased the RR of SMBC, irrespective of experimental duration, tree age, cover type, cover pattern, and residue management ($p < 0.05$) (Fig. 3). However, there were no significant SMBC changes under natural grass and mixed grass species cover and under management without mowing ($p > 0.05$). Overall, GCM increased SMQ content relative to CT, except for changes in SMQ that were not significant under experimental duration > 6 years and management without mowing ($p > 0.05$).

Response of SOC, SMBC, and SMQ to GCM in various climatic zones and latitudes

GCM significantly increased SOC (18%–24%), SMBC (20%–54%), and SMQ content (12%–27%) relative to CT (Fig. 4 and Table S5). The strongest GCM-induced increases in SMBC and SMQ content were under MAP > 1000 mm (54% and 27%, respectively), while SOC showed the highest response to GCM (24%) when MAP was between 600 and 1000 mm. Overall, the effects of GCM on SOC, SMBC, and SMQ relative to CT were positive.

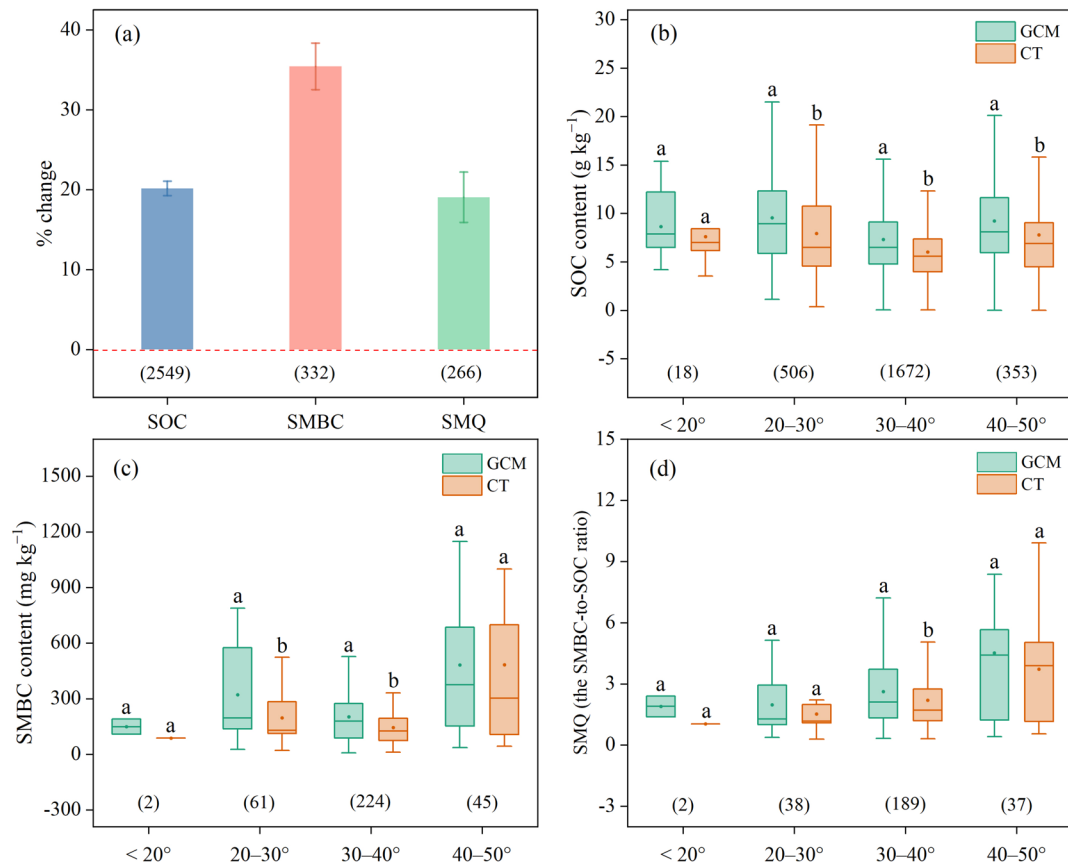


Fig. 2 Overall effect of GCM on SOC, SMBC, and SMQ in orchards **(a)**. Concentration of SOC **(b)** and SMBC **(c)**, and SMQ **(d)** in different latitudes. Numbers in parentheses represent the sample size. Error bars indicate 95% CI. If 95%

CI did not overlap with 0, it indicates significant effect sizes ($p < 0.05$). SOC, soil organic carbon; SMBC: soil microbial biomass carbon; SMQ: the ratio of SMBC to SOC

Response of SOC, SMBC, and SMQ to GCM as affected by edaphic variables

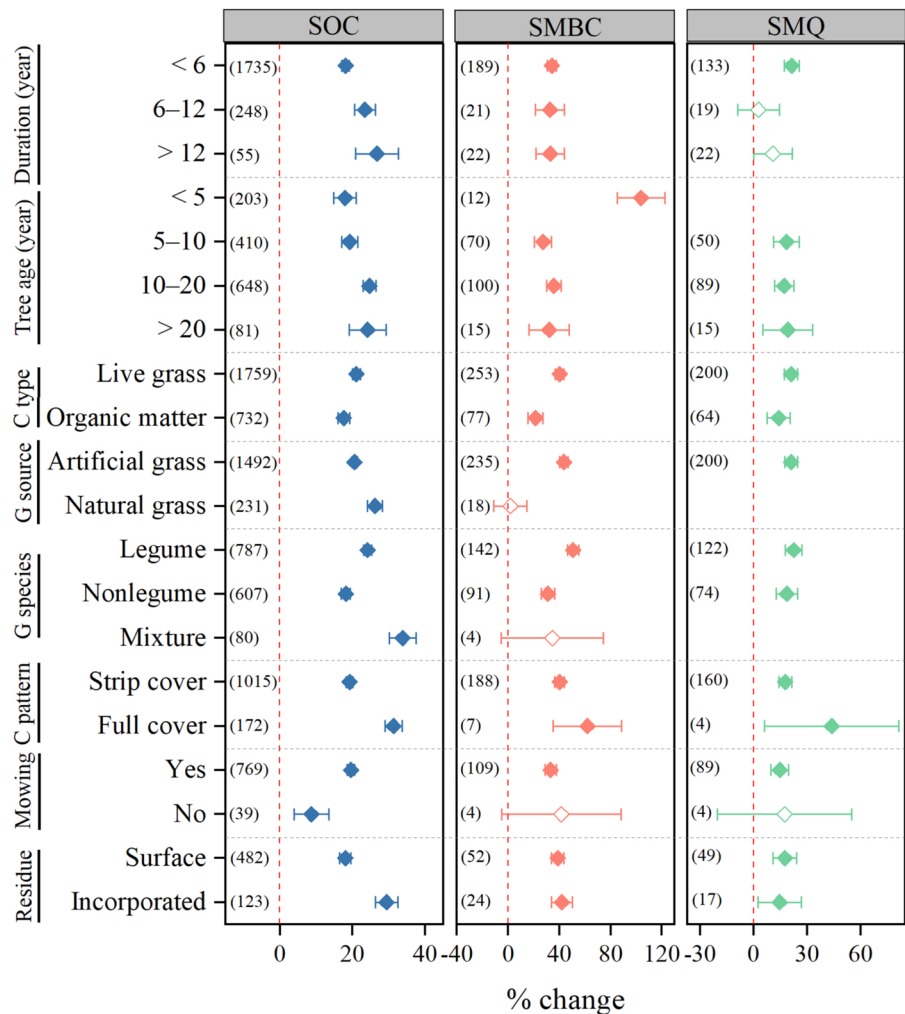
On average, GCM increased SOC (by 6%–29%), SMBC (by 5%–37%), and SMQ (by 2%–31%) (Fig. 5 and Table S6). Relative to CT, SOC content significantly increased under GCM in the 0–100 cm soil profile. SOC content in the 0–100 cm decreased with increasing soil depth. Overall, GCM enhanced SMBC content under all levels of initial SOC, TN, and pH and at soil depths < 60 cm ($p < 0.05$). However, no differences were detected at soil depth > 60 cm ($p > 0.05$). GCM significantly increased SMQ content under the following conditions: initial SOC < 10 g kg⁻¹ (16%), TN > 0.75 g kg⁻¹ (15%–23%), pH > 6 (12%–15%), and soil depth < 40 cm (17%–20%). However, these effects were not significant when

initial SOC was 10–20 g kg⁻¹, initial TN < 0.75 g kg⁻¹, initial pH < 6, and soil depth > 40 cm ($p > 0.05$).

Factors affecting the response of SOC, SMBC, and SMQ to GCM

The RF model explained that 47%–66% of the variation in SOC, SMBC, and SMQ response (Fig. 6). Soil depth was the leading factor affecting SOC, SMBC, and SMQ, with relative importance values of 64%, 43%, and 26%, respectively. In contrast, cover type and experimental duration had little influence on SOC, SMBC, and SMQ. Linear regression analysis revealed that GCM-induced effects on SOC exhibited no significant trend across experimental durations and MAT. However, the lnRR of SMBC showed a significant positive correlation with MAT and MAP. In

Fig. 3 Percentage change in SOC, SMBC, and SMQ under different agronomic factors. Numbers in parentheses represent the sample size. Error bars indicate 95% CI. If 95% CI did not overlap with 0, the effect size is considered significant ($p < 0.05$). SOC, soil organic carbon; SMBC: soil microbial biomass carbon; SMQ: the ratio of SMBC to SOC; C type: cover type; G source: grass source; G species: grass species; and C pattern: cover pattern



addition, initial SOC and TN contents showed significant negative and positive correlations, respectively, with the $\ln RR$ of SMQ. Overall, we present a summary of the effects of GCM on SOC, SMBC, and SMQ, providing mechanisms and a predictive understanding of soil carbon and microbial characteristics in response to GCM, as well as its potential drivers.

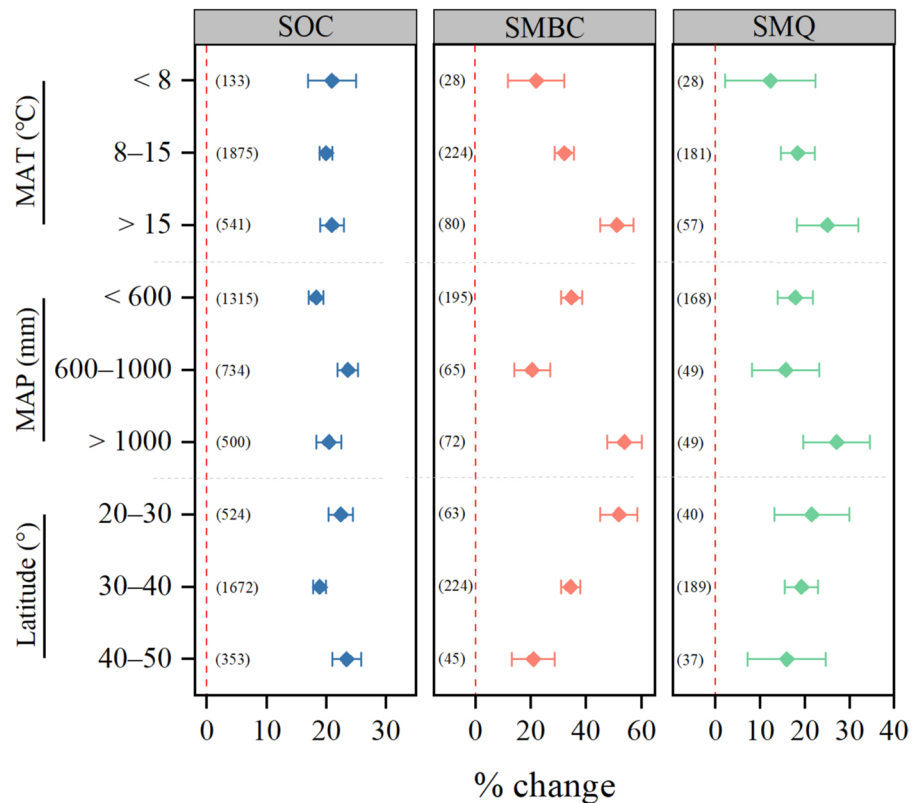
Discussion

Overall response of SOC, SMBC, and SMQ to GCM practices

Previous meta-analyses consistently demonstrated the positive effects of GCM on SOC content (Poeplau and Don 2015; Kaye and Quemada 2017; Wei

et al. 2017; Morugán-Coronado et al. 2020). Our own meta-analysis results align with these findings (Fig. 2a and Table S7), confirming our first hypothesis that GCM can bring SOC benefits (20%, $p < 0.05$). The improvement in SOC due to GCM occurs mainly by following underlying mechanisms: an increase in species composition and richness of crops under GCM practices, changing the quality of residue-derived carbon in the soil system and thus reducing SOC turnover (Hall and Russell 2019). In addition, the increased crop diversity changes the form of SOC, which increases microbial biomass and soil fertility (McDaniel et al. 2014; King and Blesh 2018), further stimulating the increase in SOC. Secondly, functional crop combination improves crop yield through niche complementarity and interspecific promotion (Ryan 2021; Fang et al. 2022). Under GCM

Fig. 4 Percentage change in SOC, SMBC, and SMQ under different climatic zones and latitudes. Numbers in parentheses represent the sample size. Error bars indicate 95% CI. If 95% CI did not overlap with 0, the effect sizes are considered significant ($p < 0.05$). SOC, soil organic carbon; SMBC: soil microbial biomass carbon; SMQ: the ratio of SMBC to SOC. MAT, mean annual temperature; MAP, mean annual precipitation

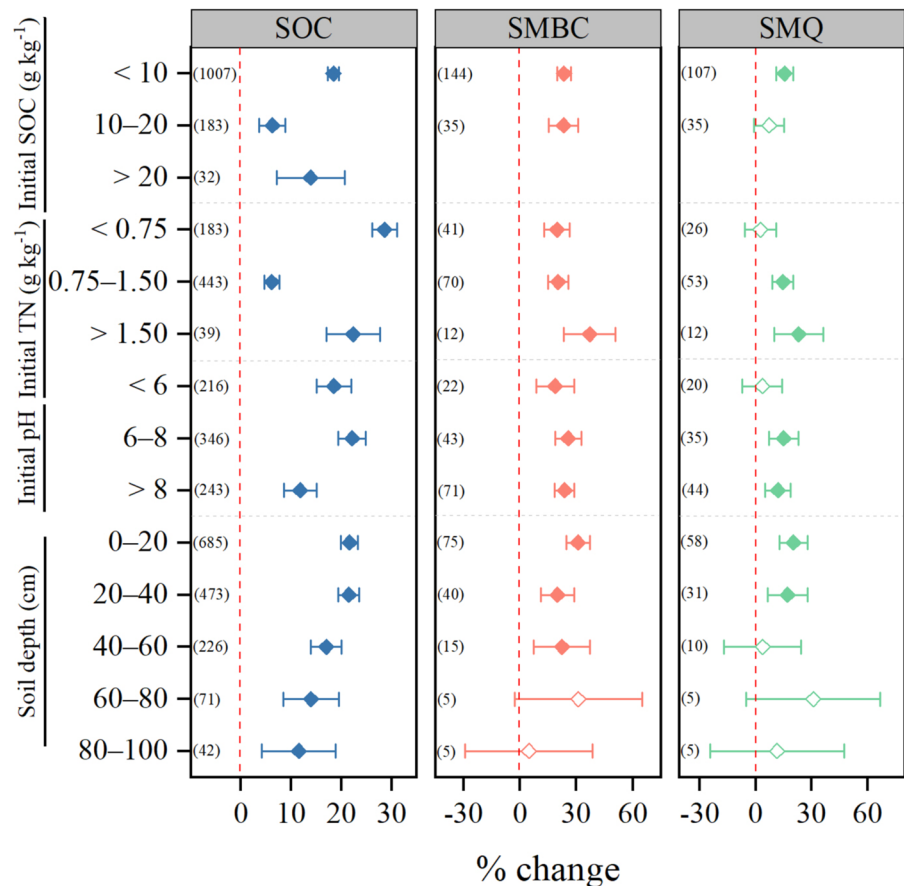


conditions, more belowground and aboveground biomass enters the soil (Pan et al. 2009). SOM consists mainly of microbial necromass and its by-products and dead cell material (Liang et al. 2017). Therefore, the increase in SOC concentration could be attributed to the increase microbial biomass (McDaniel et al. 2014). Finally, carbon input from GCM is transferred from the atmosphere to the soil through photosynthesis, thereby directly facilitating SOC accumulation (Blanco-Canqui et al. 2017). GCM increases exogenous carbon input into the system because grasses fix CO_2 -C from the atmosphere to promote their growth. The fixed carbon in the grass biomass is transferred to the soil through various processes (e.g., litter, stalk, residue) when the grass dies (Qian et al. 2014). In addition, GCM can promote organic carbon accumulation by preventing the loss of carbon via soil erosion (Zheng et al. 2021a), which eventually increases SOC concentration (Fig. 2a). The increased SMBC under GCM practice compared to CT suggests that GCM is beneficial to microbial activity (Fig. 2a), which may be attributed to organic carbon fractions

accumulation in soil during plant growth (Yang et al. 2012; Hao et al. 2022).

The relatively higher SMBC under GCM (Fig. 2a) may indicate an increased supply of carbon substrates, enhancing soil microbial communities (Zhang et al. 2022a). In general, when the soil environment is conducive to microbial physiology, the duration of microbial activity is prolonged. This enables soil microorganisms to assimilate more carbon into their biomass, consequently resulting in an increase in SMQ (Xu et al. 2014; Ghosh et al. 2020; Cao et al. 2021). The introduction of GCM significantly increased SMQ levels in the 0–40 cm soil layer (Fig. 2a). This increase can be attributed to changes in the vegetation composition and SOM content of orchards following the implementation of GCM, which affected the soil microbial biomass (Lepcha and Devi 2020). Therefore, favorable environmental conditions (e.g., greater organic matter availability, nutrients, and O_2 level) and high-quality substrate help soil microorganisms efficiently assimilate carbon into biomass (Allison et al. 2010).

Fig. 5 Percentage change in SOC, SMBC, and SMQ under different edaphic variables. Numbers in parentheses represent the sample size. Error bars indicate 95% CI. If 95% CI did not overlap with 0, the effect size is considered significant ($p < 0.05$). SOC, soil organic carbon; SMBC: soil microbial biomass carbon; SMQ: the ratio of SMBC to SOC; TN, total nitrogen



Factors driving GCM-induced changes in SOC, SMBC, and SMQ

Our findings suggested that there were highly diverse outcomes under different climatic factors, initial soil properties, and agronomic practices, which can have positive, negative, or negligible effects on SOC, SMBC, and SMQ (Figs. 3–5 and Table S8). The positive impact of GCM on SOC in mature fruit tree orchards was more pronounced than that observed in younger orchards (Fig. 3)(Xiang et al. 2022). This disparity is primarily attributed to the enhanced soil quality resulting from the prolonged application of GCM (Qian et al. 2015). Rodrigo-Comino et al. (2018) noted that soil erosion processes decrease with orchard age, which further contributes to the reduction of SOC losses. The selection of cover type, grass source, grass species, cover pattern, mowing, and residue management within the GCM system significantly influences soil carbon accumulation (Fig. 3)

(Blanco-Canqui et al. 2015; Zhao et al. 2015; King and Blesh 2018). Legume grass with full coverage, regular mowing, and the incorporation of residue into the soil contributes to the greatest improvement in SOC, SMBC, and SMQ (Fig. 3). Legumes typically have a carbon-to-nitrogen ratio of less than 20, which promotes faster decomposition and nitrogen mineralization, positively influencing microorganisms (Fustec et al. 2010; Blanco-Canqui et al. 2015). In addition, our synthetic analysis, along with other studies, has jointly demonstrated that full cover management has a greater impact on SOC, SMBC, and SMQ compared to strip cover. This difference is attributed to the coverage and biomass of full cover being twice as high as those of strip cover, as well as the increase in residual organic matter content and root penetration with greater coverage (Tu et al. 2021). Regular mowing of orchard grass, combined with the incorporation of residues into the soil, ultimately resulted in increased levels of SOC and SMBC (Fig. 3). Similarly, Oliveira

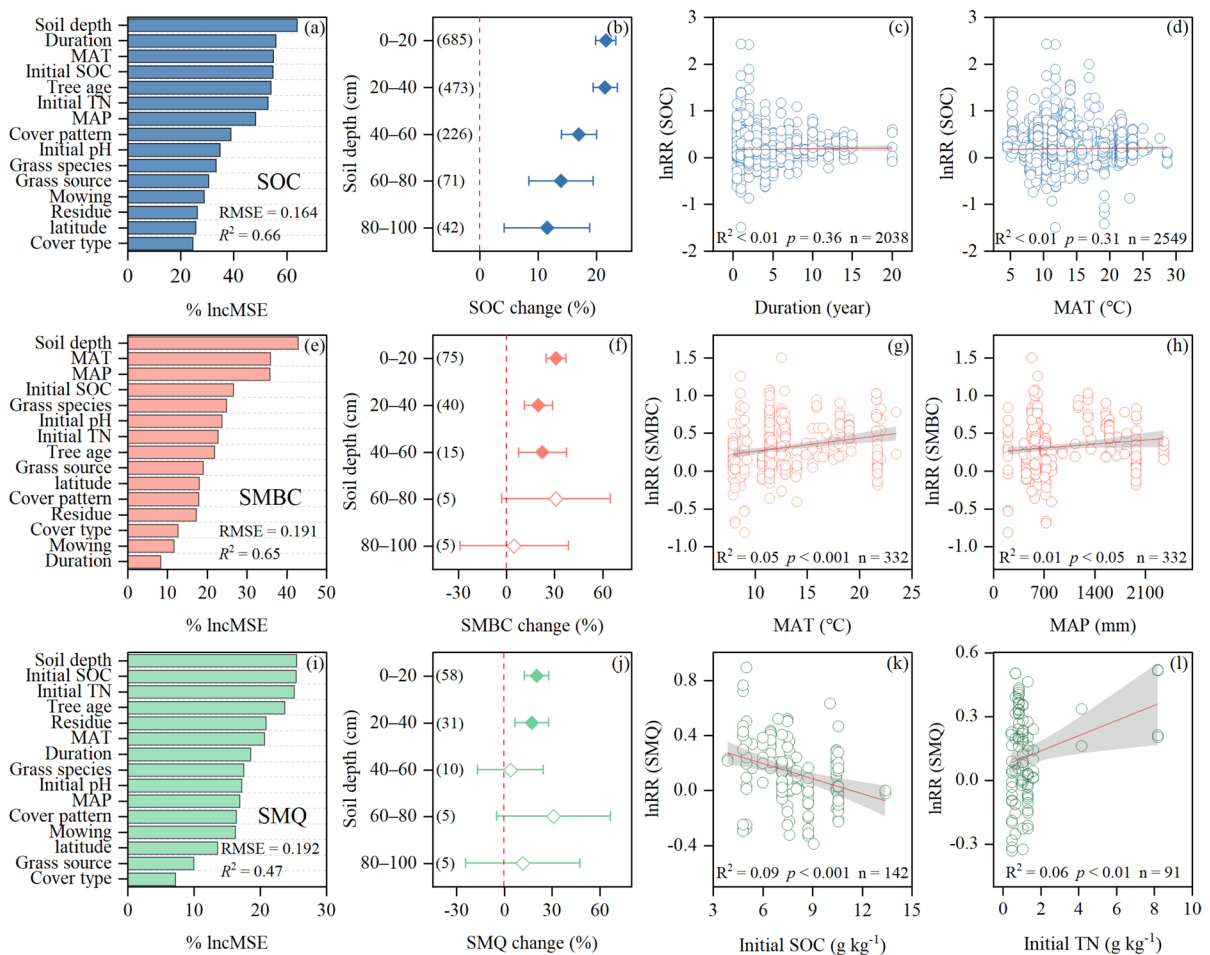


Fig. 6 The relative influence of factors on SOC, SMBC, and SMQ in the RF model and correlation analysis of SOC, SMBC, and SMQ with their top three important factors. Numbers in parentheses represent the sample size. SOC, soil

organic carbon; SMBC: soil microbial biomass carbon; SMQ: the ratio of SMBC to SOC; RF: random forest; MAT, mean annual temperature; MAP, mean annual precipitation; TN, total nitrogen

et al. (2016) found that mowing had a less detrimental effect on SOC compared to not mowing. The incorporation of grass residues into the soil leads to higher levels of SOC (Fig. 3), probably because tillage operations mix the residues throughout the soil profile, stimulating the accumulation of carbon and nitrogen (Zheng et al. 2022). Overall, the effect of GCM on SOC, SMBC, and SMQ varies with agronomic practices (Fig. 3). Therefore, selecting suitable cover crop types, grass species, cover patterns, and mowing and residue management is an effective way to maximize SOC benefits and minimize potential risks.

The contribution of GCM to SOC, SMBC, and SMQ was mainly limited by agronomic practices,

climatic conditions, and soil properties, especially soil depth according to the RF analysis (Fig. 6). We found that SOC exhibited a decrease with soil depth, and this pattern is similar to what has been reported in previous publications (Adrover et al. 2017; Sun et al. 2020; Liu et al. 2022; Lv et al. 2023). The significant change in GCM-induced SOC content (0–100 cm) with decreasing soil depth may be associated with improvements in organic matter input, microbial biomass, and soil structure in the topsoil (Badagliacca et al. 2021; Lv et al. 2023). The impact of GCM on SOC in the topsoil was significantly more pronounced than in the subsoil (Fig. 5). Crop residues remained in the surface soil, while there was clear stratification

of soil carbon due to decreasing depth into the deeper soil (Li et al. 2021a).

GCM-induced SMBC content decreased with soil depth (Fig. 6). This can be attributed to the higher abundance of surface soil substrate and nutrient inputs from GCM residue compared to the subsoil. Additionally, the living conditions, such as O₂ levels, are more favorable for the growth of surface microorganisms (Jiao et al. 2018; Sun et al. 2020). The overall decrease in SMQ with increasing soil depth indicates that microorganisms have a reduced ability to assimilate soil carbon as the soil depth increases (Fig. 6). The topsoil has greater organic matter inputs, soil macro-aggregation, and microbial biomass (Badagliacca et al. 2021). Substrate and nutrients play a major role in controlling microbial growth (Sun et al. 2020). Secondly, plant root distribution decreases with soil depth, which directly leads to a reduction in root exudates and rhizosphere conduits, thus limiting the energy source supply necessary for microbial growth (Yan et al. 2022), and ultimately leads to a decrease in microbial entropy with soil depth. Thirdly, changes in soil conditions such as temperature, water availability, compaction/porosity, alkalinization, and oxygen levels at different soil depth may also lead to a decrease in SMQ (Malik et al. 2018). Finally, numerous studies have highlighted the higher sensitivity of SMBC, compared to SOC, to variations in climate, agronomic practices, and soil factors (Sun et al. 2020; Li et al. 2021c; Muhammad et al. 2021). The collective findings indicate that SMBC declines more rapidly with increasing soil depth than SOC (Fig. 4), indicating that SMQ is more responsive to both biological and abiotic changes.

Experimental duration was identified as the second vital driver affecting GCM-induced SOC dynamics at regional scales (Fig. 6). We observed that the responses of SOC to GCM increased significantly with experimental duration (Figs. 3–6). This finding aligns with previous studies that reported a positive correlation between SOC accumulation and the duration since GCM was introduced in orchards, likely due to the sustained input of organic matter over time (Wei et al. 2017, 2018; Xiang et al. 2022). Climatic factors (i.e., MAT and MAP) ranked second and third in importance for SOC and SMBC content in response to GCM practice (Figs. 4–6). GCM significantly increased SOC and SMBC content across all climate types. The significant positive effects

of GCM in relatively wet and warm climate conditions can be attributed to the higher levels of gross primary production typically found in these regions. This increased production enhances both below- and above-ground biomass inputs into the soil, ultimately stimulating the accumulation of SOC and SMBC (Barman et al. 2014). Initial SOC and TN contents had significant effects on the response of SMQ to GCM practices (Figs. 5 and 6). Our findings indicate that GCM was more effective in enhancing SMQ in soils with low initial SOC content and high levels of initial TN content (Figs. 5 and 6). In general, a high initial SOC concentration is anticipated to result in a lower rate of SOC accumulation, whereas soils with low SOC concentrations prior to GCM are more likely to exhibit a significant carbon saturation deficit (Powlson et al. 2011; Vicente-Vicente et al. 2017). It is well known that higher TN content stimulates crop growth, which in turn leads to greater enhancement of microbial biomass in the soil (Liu et al. 2022). Overall, given that SOC, SMBC, and SMQ induced by GCM practices are quite complex due to variations in climatic, edaphic, and agronomic factors, conducting field-level cover crop suitability assessments is essential for developing site-specific strategies to enhance SOC under GCM.

Implications, limitations, and prospects

Our meta-analysis, based on 3147 paired data from 288 field experiment sites (Fig. 1), quantitatively evaluated changes in SOC, SMBC, and SMQ influenced by GCM practices and associated climate, edaphic factors, and agronomic practices. We found that, on average, GCM significantly increased SOC, SMBC, and SMQ (Fig. 2a). The highly diverse outcomes under different management and environmental factors, with positive, negative or nil effects on SOC, SMBC, and SMQ, help explain past uncertainties and disagreement regarding the potential benefits of adopting GCM practices (Figs. 3–5). The analysis presented herein indicated soil properties (soil depth) control the variations in SOC, SMBC, and SMQ (Fig. 6). Overall, the results generally suggested that adopting GCM practices could improve the carbon sink and boost the biological soil health in Chinese orchards. We also emphasize the importance of considering local environmental factors

when establishing and managing site-specific GCM systems for sustainable agricultural production.

Uncertainty remains in the present study on the effect size of GCM on SOC, SMBC, and SMQ based on the meta-analysis. Data limitations hindered us from examining other known important factors, such as soil texture, soil type, root structure, grass age, tillage intensity, cropping intensity, crop species diversity, irrigation, fertilizer management, and cropping systems. These factors affect the quantity and quality of soil carbon inputs to varying degrees, thereby affecting the soil microbial activity (Li et al. 2017) and ultimately changing SMBC dynamics and SOC turnover (Liu et al. 2014). Most of the published studies reported variations in topsoil SMBC and SMQ, and some studies lacked SD values. This study estimated SD values based on existing data. A number of factors contribute to additional uncertainties in the data set.

Given that microorganisms both assimilate carbon and release CO₂ through respiration, it is critical to consider priming effects or SOC losses. Collectively, the inclusion of more potential variables (e.g., plant traits) and key indicators (e.g., microbial carbon pump efficiency) will help deepen our understanding of GCM-induced changes in soil carbon cycling. Additionally, to improve the accuracy of the outcomes, more detailed and rigorous field experiments are needed to verify the effects of GCM on soil microbiological characteristics and soil carbon accumulation. Therefore, subsequent studies should focus on a comprehensive quantitative synthesis of the effects of GCM on microbial strategies.

Conclusion

In this meta-analysis, we provided a comprehensive and systematic assessment of the impacts of GCM on SOC, SMBC, and SMQ in orchards across China. These results confirm that GCM significantly increased SOC (20%), SMBC (35%), and SMQ (19%) relative to continuous CT in orchards. Climatic, edaphic, and agronomic practices significantly influenced the responses of SOC, SMBC, and SMQ to GCM. Effects in SOC, SMBC, and SMQ under GCM relative to CT were mainly controlled by soil properties and climatic conditions. Soil depth was the primary driving factor for the response of SOC,

SMBC, and SMQ to GCM practice. In summary, our study provides deeper insights into the identification of site-specific GCM practices that enhance SOC in agroecosystems and ultimately contribute to carbon neutrality.

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Author contribution Weiting Ding: Visualization, Writing-original draft, Writing-re-view & editing. Francis Zvomuya: Writing-review & editing. Yeru Wu, Jinmei Chen, Jianuo Mu, Zhidong Qi: Writing-review & editing. Hailong He: Conceptualization, Writing-review & editing.

Data availability Data will be made available on reasonable request.

Declarations

Competing interest The authors declare that they have no known competing interests.

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